

Name: Matsuda

Student ID: \_\_\_\_\_

1.

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Vector values are ( a, b, c, d, g, h, i )

2.

10

$$(a) \quad \frac{d}{dt}(r \cos \theta) = \dot{r} \cos \theta - r \sin \theta \dot{\theta}$$

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$$(b) \quad \frac{d}{dt}(P) = (\dot{r} \cos \theta - r \sin \theta \dot{\theta}) \hat{i} \\ + (\dot{r} \sin \theta + r \cos \theta \dot{\theta}) \hat{j}$$

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3.

$$(15) \quad (a) \quad m\ddot{x} = mg - cx$$

$$(15) \quad (b) \quad m\dot{v} + cv = mg$$

$$m \frac{dv}{dt} = mg - cv$$

$$\frac{m}{mg - cv} dv = dt$$

$$\frac{1}{v - \frac{mg}{c}} dv = -\frac{c}{m} dt$$

$$\ln \left( v - \frac{mg}{c} \right) = -\frac{c}{m} t + A$$

$$v - \frac{mg}{c} = A' \exp \left( -\frac{c}{m} t \right)$$

$$v = A' \exp \left( -\frac{c}{m} t \right) + \frac{mg}{c}$$

$$x = \int_0^t v dt = \left[ A' \left( -\frac{m}{c} \right) \exp \left( -\frac{c}{m} t \right) + \frac{mg}{c} t \right]$$

$$= -A' \left( \frac{m}{c} \right) \left\{ \exp \left( -\frac{c}{m} t \right) - 1 \right\} + \frac{mg}{c} t + D.$$

$$v|_{t=0} = A' + \frac{mg}{c} = 0 \quad A' = -\frac{mg}{c}$$

$$x|_{t=0} = D = 0 \quad D = 0$$

$$x = g \frac{m^2}{c^2} \left( \exp \left( -\frac{c}{m} t \right) - 1 \right) + \frac{mg}{c} t.$$

$$v = -\frac{mg}{c} \exp \left( -\frac{c}{m} t \right) + \frac{mg}{c}$$

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4.

$$\boxed{15} \quad (a) \quad m\ddot{x} + kx = F(x)$$

$$\boxed{10} \quad (b) \quad x = A \sin \omega t + B \cos \omega t \quad \omega = \sqrt{\frac{k}{m}}$$

$$x|_{t=0} = B = x_0$$

$$v = \omega A \cos \omega t - \omega B \sin \omega t$$

$$v|_{t=0} = \omega A = 0$$

$$x = x_0 \cos \sqrt{\frac{k}{m}} t$$

$$\boxed{5} \quad (c) \quad \text{Particular solution } x_2 = D \sin \omega_0 t \Rightarrow m\ddot{x} + kx = F_0 \sin \omega_0 t$$

$$-m\omega_0^2 D \sin \omega_0 t + k D \sin \omega_0 t = F_0 \sin \omega_0 t$$

$$D(k - m\omega_0^2) \sin \omega_0 t = F_0 \sin \omega_0 t$$

$$D = \frac{F_0}{k - m\omega_0^2}$$

General solution is

$$x = A \sin \omega t + B \cos \omega t + \frac{F_0}{k - m\omega_0^2} \sin \omega_0 t$$

$$x|_{t=0} = B = 0$$

$$v = \omega A \cos \omega t - \omega B \sin \omega t + \omega_0 \frac{F_0}{k - m\omega_0^2} \cos \omega_0 t$$

$$v|_{t=0} = \omega A + \frac{F_0 \omega_0}{k - m\omega_0^2} = v_0 \quad A = \frac{1}{\omega} \left( v_0 - \frac{F_0 \omega_0}{k - m\omega_0^2} \right)$$

$$x = \sqrt{\frac{m}{k}} \left( v_0 - \frac{F_0 \omega_0}{k - m\omega_0^2} \right) \sin \omega t + \frac{F_0}{k - m\omega_0^2} \sin \omega_0 t$$